

Online Appendix

Chained Indices Unchained:

Structural Transformation and the Welfare Foundations of Income Growth Measurement

Omar Licandro

Juan Ignacio Vizcaino

U. of Leicester and BSE

University of Nottingham

April 2026

1 Introduction

This note extends the benchmark structural transformation model of [Herrendorf et al. \(2021\)](#), hereafter HRV (2021), by introducing sector-specific wedges in the transformation of production output into usable consumption goods.

The benchmark model implies that relative prices are determined entirely by relative sectoral productivities. In contrast, we allow the prices relevant for production decisions to differ from the prices relevant for household consumption decisions.

Let P_{jt} denote the *production price* of sector $j \in \{g, s\}$, which is the price relevant for firms' production decisions. Furthermore, let P_{jt}^τ denote the *expenditure price* faced by households when purchasing usable consumption units. The wedge between these prices captures the fact that delivering one unit of usable consumption requires more than one unit of raw output.

Importantly, as we show below, this wedge does not affect firms' production decisions and therefore does not contaminate the measurement of sectoral productivities. Production

technologies, preferences, and the investment aggregator remain identical to those in HRV (2021).

2 Transformation Technology

Let $\tau_j \in \mathbb{R}$ denote a sector-specific transformation parameter for sector $j \in \{g, s\}$.

Delivering one unit of usable consumption requires $e^{\tau_j t}$ units of raw output. The relationship between production prices and expenditure prices is therefore

$$P_{jt}^\tau = e^{\tau_j t} P_{jt}, \quad j \in \{g, s\}. \quad (1)$$

When $\tau_j = 0$, the benchmark HRV (2021) model is recovered. The investment sector purchases raw output at production prices P_{jt} and is therefore unaffected by the transformation wedge.

3 Representative Firms' Problems

Sectoral Production. Production technologies are identical to HRV (2021)

$$Y_{jt} = A_{jt} K_{jt}^\theta L_{jt}^{1-\theta}. \quad (2)$$

The representative firm solves

$$\max_{K_{jt}, L_{jt}} P_{jt} A_{jt} K_{jt}^\theta L_{jt}^{1-\theta} - R_t K_{jt} - W_t L_{jt}. \quad (3)$$

First-order conditions are

$$\theta P_{jt} A_{jt} K_{jt}^{\theta-1} L_{jt}^{1-\theta} = R_t \quad (4)$$

$$(1 - \theta) P_{jt} A_{jt} K_{jt}^\theta L_{jt}^{-\theta} = W_t. \quad (5)$$

Taking ratios,

$$\frac{K_{jt}}{L_{jt}} = \frac{\theta}{1-\theta} \frac{W_t}{R_t}. \quad (6)$$

Hence the capital-labor ratio is equalized across sectors.

With total labor normalized to one,

$$\frac{K_{gt}}{L_{gt}} = \frac{K_{st}}{L_{st}} = K_t. \quad (7)$$

As shown below, because the capital-labor ratio is equalized across sectors, the **key aggregation result** of the benchmark model continues to hold.

Relative Production Prices. Using (4)–(7) we obtain

$$\frac{P_{gt}}{P_{st}} = \frac{A_{st}}{A_{gt}}. \quad (8)$$

Relative production prices are therefore determined entirely by relative sectoral productivities, as in HRV (2021).

Using (1), the relative expenditure price becomes

$$\frac{P_{gt}^\tau}{P_{st}^\tau} = \frac{A_{st}}{A_{gt}} e^{-(\tau_s - \tau_g)t}. \quad (9)$$

Thus the wedge affects structural transformation only through the prices relevant for household expenditure.

4 Investment Block

The investment technology is identical to HRV (2021):

$$1 = \frac{[\omega_x P_{gt}^{1-\varepsilon_x} + (1-\omega_x) P_{st}^{1-\varepsilon_x}]^{\frac{1}{1-\varepsilon_x}}}{A_{xt}}. \quad (10)$$

Solving for sectoral prices yields

$$P_{jt} = \frac{\mathcal{A}_t}{A_{jt}}, \quad j \in \{g, s\}. \quad (11)$$

where

$$\mathcal{A}_t = A_{xt} \left[\omega_x A_{gt}^{\varepsilon_x - 1} + (1 - \omega_x) A_{st}^{\varepsilon_x - 1} \right]^{\frac{1}{\varepsilon_x - 1}}. \quad (12)$$

Expenditure prices follow from (1):

$$P_{jt}^\tau = e^{\tau_{jt}} \frac{\mathcal{A}_t}{A_{jt}}. \quad (13)$$

5 Aggregation

Using (11), (2) and (7), we get

$$P_{jt} Y_{jt} = \mathcal{A}_t K_t^\theta L_{jt}. \quad (14)$$

Summing across sectors yields

$$Y_t = \mathcal{A}_t K_t^\theta. \quad (15)$$

Hence the aggregate production representation derived in HRV (2021) remains unchanged. Factor prices are

$$R_t = \theta \mathcal{A}_t K_t^{\theta - 1} \quad (16)$$

$$W_t = (1 - \theta) \mathcal{A}_t K_t^\theta. \quad (17)$$

6 Representative Household

Households purchase usable consumption units at expenditure prices P_{jt}^τ .

Total expenditure is

$$E_t^\tau = P_{gt}^\tau C_{gt} + P_{st}^\tau C_{st}. \quad (18)$$

7 Resource Constraint

The transformation technology implies

$$Y_{jt} = X_{jt} + e^{\tau_{jt}} C_{jt}. \quad (19)$$

Multiplying by P_{jt} and summing across sectors gives

$$Y_t = X_t + E_t^\tau. \quad (20)$$

Thus aggregate output measured at production prices equals investment plus household expenditure measured at expenditure prices.

8 Balanced Growth Path

Since production and investment operate at production prices, the aggregate balanced growth path (ABGP) is governed by the benchmark effective productivity $\mathcal{A}_t$, exactly as in the benchmark HRV (2021) model. Defining growth rates using hat notation, we get that the rental rate of capital is constant along the ABGP if

$$\hat{K}_t = \frac{\hat{\mathcal{A}}_t}{1 - \theta}.$$

This, together with (15) yields

$$\hat{Y}_t = \hat{\mathcal{A}}_t + \theta \hat{K}_t, \quad (21)$$

The law of motion for capital implies X_t/K_t is constant, so $\hat{X}_t = \hat{K}_t$. This, together with (20) yields $\hat{E}_t^\tau = \hat{K}_t$. It follows that

$$\hat{Y}_t = \hat{X}_t = \hat{E}_t^\tau. \quad (22)$$

Therefore, the wedge does not alter the growth rates of aggregate variables. In particular, the laws of motion for Y_t , K_t , X_t , R_t , and W_t remain exactly as in HRV (2021), with $\hat{E}_t^\tau$ now growing at the same rate as the remaining aggregate variables.

Finally, given an initial capital stock K_0 , the initial level of consumer expenditure that puts the system on the ABGP is

$$E_0^\tau = K_0 \left[\mathcal{A}_0 K_0^{(\theta-1)} - \hat{K} - \delta \right]. \quad (23)$$

Equation (23) shows that, conditional on the production-side allocation, the wedge affects the initial level of consumption expenditure required for the model to be on an ABGP from the first period, but it does not change the growth rates of the aggregate objects along the ABGP.

9 Measurement

The model distinguishes between two price concepts. Production prices P_{jt} correspond to the prices relevant for firms' production decisions and for the measurement of value added. Expenditure prices P_{jt}^τ correspond to the prices faced by households when purchasing usable consumption units.

In empirical implementation:

- Sectoral productivities A_{jt} are recovered from production data using sectoral real value added and input quantities,

$$A_{jt} = \frac{Y_{jt}}{K_{jt}^\theta L_{jt}^{1-\theta}}. \quad (24)$$

- Aggregate productivity is recovered from the Solow residual using aggregate value added deflated by the investment price,

$$\mathcal{A}_t = \frac{Y_t}{K_t^\theta}. \quad (25)$$

- Observed relative prices correspond to the expenditure prices P_{jt}^τ .
- We normalize the service wedge to zero,

$$\tau_s = 0, \quad (26)$$

and define

$$\zeta := \tau_s - \tau_g = -\tau_g. \quad (27)$$

- To estimate ζ , we proceed in three steps. First, we compute the relative price of goods to services in the data, $P_{g,t}^\tau/P_{s,t}^\tau$. Second, we construct the corresponding relative price implied by the benchmark HRV (2021) model without the wedge, given by

$$\frac{P_{g,t}}{P_{s,t}} = \frac{A_{s,t}}{A_{g,t}}. \quad (28)$$

Third, we take the ratio of the data relative price to the model-implied relative price,

$$\frac{P_{g,t}^\tau/P_{s,t}^\tau}{A_{s,t}/A_{g,t}}, \quad (29)$$

and fit an exponential trend to this series. From (9), this ratio evolves according to

$$\frac{P_{g,t}^\tau/P_{s,t}^\tau}{A_{s,t}/A_{g,t}} = e^{-\zeta t}. \quad (30)$$

Estimating the exponential trend therefore provides a direct estimate of ζ . This procedure yields

$$\zeta = 0.0137. \quad (31)$$

This procedure isolates the component of relative price movements that cannot be accounted for by sectoral productivity differences and attributes it to the transformation wedge.

References

Herrendorf, B., Rogerson, R., and Valentinyi, A. (2021). Structural change in investment and consumption—a unified analysis. *The Review of Economic Studies*, 88(3):1311–1346.